

# **Generation Interconnection Combined Feasibility / System Impact Study Report**

## **For Queue Project Q17-0766**

Eaton 100 kV  
October 15, 2018

### ***1. Introduction***

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0766. The Interconnection Customer proposes a 284.8 MW natural gas combined cycle facility interconnecting at the Eaton 100 kV substation in the Indiana area. The request was received on March 25, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 287 higher-queued active requests and their assigned network upgrades.

### ***2. Summary of Results***

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$26,573,000.

### ***3. Network Impacts***

#### ***3.1 Generator Deliverability and Single Contingency Analysis***

1. Facility: Cape North – Eaton 100 kV  
Contingency: Headwaters Wind Farm Li Llc – Eaton 100 kV  
Loading: 129.2% (Rating: 135 MVA)  
DFAX: 53.8%  
Pre-project loading: 15.3%
2. Facility: Cape North – Anchor Hocking 100 kV  
Contingency: Headwaters Wind Farm Li Llc – Eaton Substation 100 kV  
Loading: 126.2% (Rating: 131 MVA)  
DFAX: 53.8%  
Pre-project loading: 9.5%
3. Facility: Headwaters Wind Farm Li Llc Substation – Eaton 100 kV  
Contingency: Eaton – Cape N 100 kV line  
Loading: 123.8% (Rating: 137 MVA)  
DFAX: 66.2%  
Pre-project loading: 13.7%
4. Facility: Headwaters Wind Farm Li Llc – Murock 100 kV line  
Contingency: Eaton – Cape N 100 kV  
Loading: 118.2% (Rating: 139 MVA)  
DFAX: 66.2%  
Pre-project loading: 17.5%
5. Facility: Canal St – Murdock 100 kV Ckt 1  
Contingency: Canal St – Murdock 100 kV  
Loading: 113.8% (Rating: 151 MVA)  
DFAX: 8.0%  
Pre-project loading: 98.7%

6. Facility: Canal T – Murdock 100 kV Circuit 2  
 Contingency: Mrdock – Canal St 100 kV #1  
 Loading: 110.2% (Rating: 156 MVA)  
 DFAX: 8.0%  
 Pre-project loading: 95.7%
7. Facility: Omni-Hackberry 500/230kV Transformer Bank  
 Contingency: Omni-Hackberry – Norwich 500 kV  
 Loading: 107.0% (Rating: 942 MVA)  
 DFAX: 8.1%  
 Pre-project loading: 104.5%

### **3.2 Multiple Facility Contingency Analysis**

No additional overloads were identified for multiple facility contingencies.

### **3.3 Short Circuit and Stability**

Short circuit and stability analyses identified no additional network upgrades.

## **4. Required Network Upgrades and Cost Estimates**

| Facility                                         | Required Upgrade                            | Cost Estimate | Allocation to Project |
|--------------------------------------------------|---------------------------------------------|---------------|-----------------------|
| Cape North – Eaton 100 kV                        | Rebuild line with higher-capacity conductor | \$4,295,000   | 100% (\$4,295,000)    |
| Cape North – Anchor Hocking 100 kV               | Rebuild line with higher-capacity conductor | \$2,922,000   | 100% (\$2,922,000)    |
| Headwaters Wind Farm li Llc – Eaton 100 kV       | Rebuild line with higher-capacity conductor | \$5,330,000   | 100% (\$5,330,000)    |
| Headwaters Wind Farm li Llc – Murock 100 kV line | Rebuild line with higher-capacity conductor | \$12,282,000  | 100% (\$12,282,000)   |
| Murdock – Canal St Substation 100 kV L           | Rebuild line with higher-capacity conductor | \$1,534,000   | 19% (\$292,000)       |
| Murdock – Canal St 100 kV Cir. 2                 | Rebuild line with higher-capacity conductor | \$1,137,000   | 19% (\$216,000)       |
| Omni-Hackberry 500/230kV Transformer Bank        | Replace transformer with higher-rated unit  | \$35,281,000  | 4% (\$1,236,000)      |
|                                                  | Total allocated to project                  |               | \$26,573,000          |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.